Seeing Through Solid Rock

MYSSP 2026 demo-day deck: 12-page specification and full script

Evan Lim Hong Yan Junyi Chai Samantha Chan Teng Wei

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Deck map and production rules

12 pages total: page 1 title; pages 2–11 main content; page 12 thank-you and references. The live slot is 15 minutes plus 10 minutes Q&A, so each content slide gets about 80 seconds. Body text is 20–28 pt, titles 36–44 pt, and nothing is below 16 pt. Each page has one dominant visual and at most five short bullets; the script carries the detail. All numbers come from `figures/results_summary.json` and the report. Slide 6 and 8 are the only pages that need a live/recorded notebook run; the rest are static.

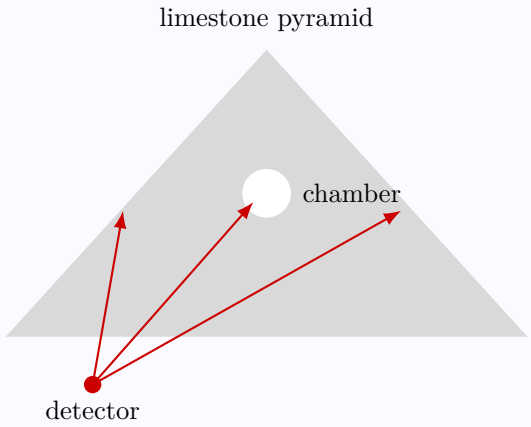
Page	Content	Visual
1	Title and headline	TikZ logo
2	Why muons can see through rock	atmosphere + energy-loss diagram
3	Muography, question, baseline	ray diagram + parameter table
4	Five-step method	pipeline flowchart
5	Ray tracing and opacity	density slice + camera-grid diagram
6	Counts and reconstruction	expected counts + clean ratio
7	Noise, significance, look-elsewhere	Poisson diagram + null figure
8	Exposure curve and headline	exposure curve + definitions table
9	Studies A and B	size and depth figures
10	Studies C and D, Malaysian target	placement, binning, hill figures
11	Systematics, limitations, conclusion	density figure + three-pillar diagram
12	Thank you, references, Q&A	reference list

Slide 1 — Title and the one-number answer 0:40

On the slide.

- **Seeing Through Solid Rock**
- Finding a hidden chamber with cosmic-ray muons
- **13.4 days** for the expected signal to reach 5σ (3σ at 4.8 days)
- MYSSP 2026 · Evan Lim, Junyi Chai, Samantha Chan

Figure / diagram. a clean TikZ logo: pyramid, chamber, detector, three rays; one large number in the corner.



13.4 days

for 5σ

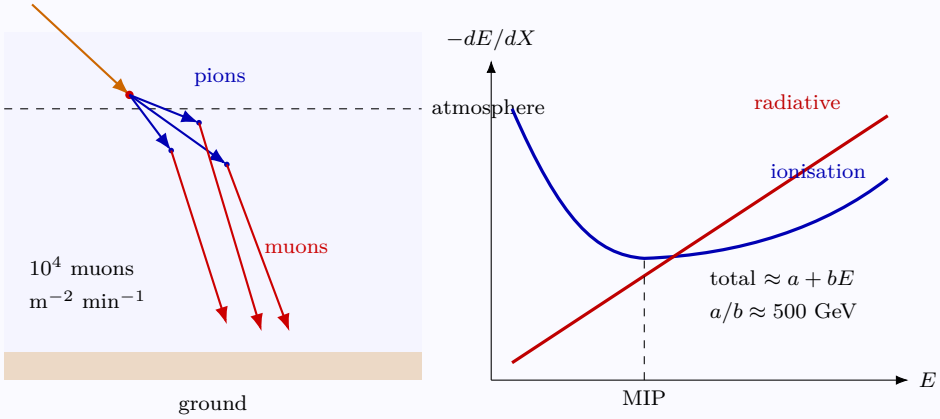
Spoken script. Good morning. Our project asks how long a small muon detector must run before it can see a hidden chamber inside a solid limestone pyramid. The answer under our baseline model is 13.4 days for a five-sigma expected signal, and 4.8 days for three sigma. I will show why muons can do this, how the simulation works, how we separate signal from noise, and what changes the answer.

Slide 2 — Why muons can see through rock 1:20

On the slide.

- Cosmic-ray protons make pions; pions decay into muons.
- Muon: heavy electron, $207 m_e$, lifetime 2.2 μs, dilated by γ.
- Flux: 10^4 m^{-2} min^{-1} at sea level.
- Energy loss: $a + bE$; 2 MeV per g/cm^2; 100 m limestone needs 52.6 GeV.
- Muons survive because radiative loss scales as $1/m^2$.

Figure / diagram. two large TikZ panels: (left) atmosphere shower and muon paths; (right) ionisation and radiative energy-loss curves with the MIP marked.



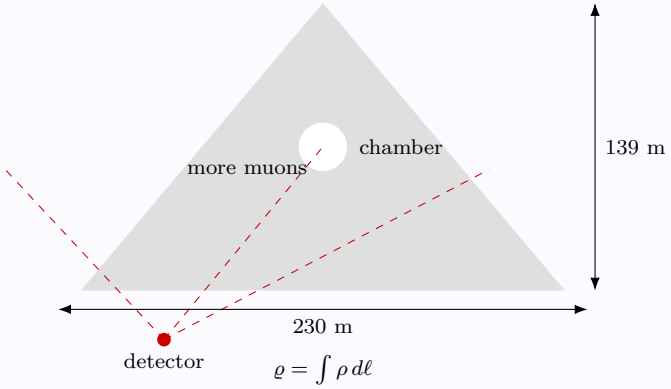
Spoken script. A cosmic-ray proton hits a nucleus about 15 kilometres up and produces pions, which decay into muons. A muon is a heavy electron: 207 times the electron mass, same charge, two-point-two microseconds at rest. Naively it decays after 660 metres, but it reaches the ground because time is dilated by gamma. Roughly ten thousand muons cross each square metre every minute. Why do they penetrate? Energy loss has two pieces: ionisation, about 2 MeV per gram per square centimetre, and radiative loss, which grows with energy. Radiative loss scales as one over mass squared, so a muon radiates about 43,000 times less readily than an electron. One hundred metres of limestone needs 52.6 GeV. Only the high-energy tail gets through, and that tail is our beam.

Slide 3 — Muography, the question, and the baseline 1:20

On the slide.

- Opacity $\varrho = \int \rho \, d\ell$: less rock $\rightarrow$ more muons.
- One direction = one pixel; a 2D radiograph, not a 3D CT.
- Baseline: 230 m base, 139 m high, 10 m chamber, 0.25 m^2 detector, $\pm 40°$, $2°$ pixels.
- Question: how many days for the *expected* signal to reach 5σ?

Figure / diagram. left: large TikZ ray diagram through the pyramid; right: a compact parameter table with the same numbers as the report.



Parameter	Value
Base / height	230 / 139 m
Chamber radius	10 m
Chamber centre	$(0, 0, 70)$ m
Detector area	0.25 m^2
Detector position	$(0, -40, 3)$ m
Field / bin	$\pm 40°$ / $2°$
Pixels	1681
Ray step	0.25 m
Seed	0

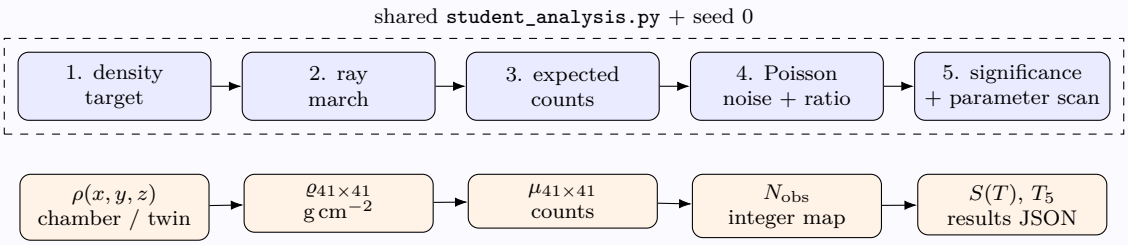
Spoken script. Muography turns that beam into an image. Along each direction we integrate the density: that is the opacity. More rock means fewer surviving muons, so a void shows up as an excess. One direction becomes one pixel; this is a radiograph, not a full three-dimensional tomography. Our baseline is a 230 metre square pyramid, 139 metres high, with a 10 metre spherical chamber at height 70. The detector is a quarter square metre at minus 40 metres, with a plus or minus 40 degree field and two degree pixels, 1681 directions. The research question is the number on the previous slide: how many days before the expected signal reaches five sigma?

Slide 4 — The five-step method, and the data it produces 1:25

On the slide.

- Density field → ray march → expected counts → noise/ratio → significance/scan.
- Two targets: chamber and filled-solid twin; their ratio removes the pyramid.
- Shared module `opacity_map`, `expected_counts`; seed 0; fresh kernel.
- Data: 1681-pixel maps, 500 null runs, 500 void runs, six one-parameter scans.

Figure / diagram. a large horizontal pipeline with icons and a data-structure inset showing the 41×41 map, the angle grids, and the two twin branches.



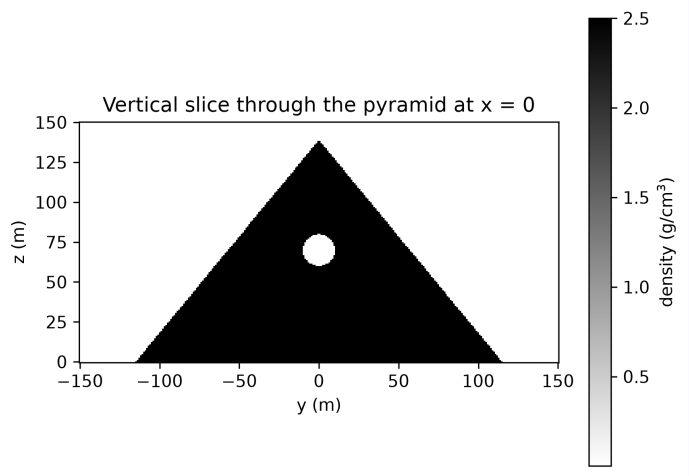
Spoken script. The method is five steps. Step one is a density function: rock, air, or a chamber. Step two is pure geometry: march a ray through every direction and add density times length. Step three converts opacity to a minimum energy and then to expected muon counts. Step four draws Poisson noise and divides by the no-void twin. Step five builds the significance map and scans one parameter at a time. The important engineering choice is that steps two and three live in a shared module, not in a notebook; every notebook imports the same functions and uses seed zero. The data products are the 1681-pixel maps, 500 null runs, 500 chamber runs, and six one-parameter scans.

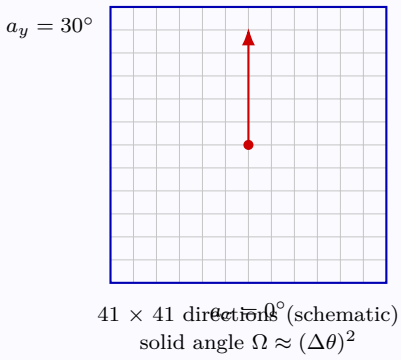
Slide 5 — Ray tracing: from geometry to an opacity map 1:20

On the slide.

- Tangent-plane camera; 1681 rays; 0.25 m steps.
- Chamber notch: 4,935 g/cm^2, against the geometric 5,000.
- Ray-step halving changes the map by $\leq 0.24\%$.
- No-void control is zero away from the chamber.

Figure / diagram. left: the density slice; right: a TikZ camera-grid showing the 41×41 directions and one highlighted signal ray at 30.8°.





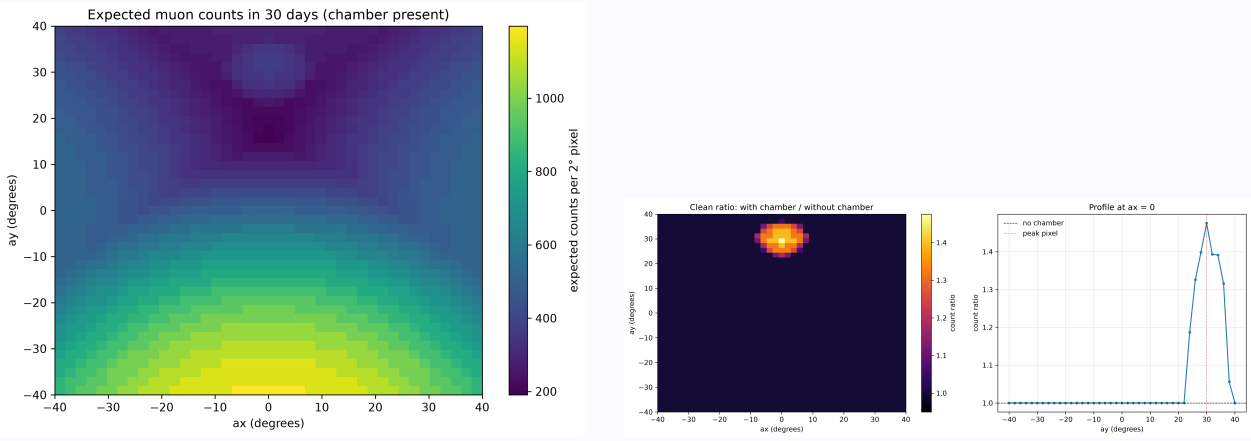
Spoken script. This is the geometry step. The camera is a tangent plane, so a direction is two small angles. We trace 1681 rays in quarter-metre steps and add density times length. The chamber is a 10 metre sphere, so the longest chord is 20 metres and the missing opacity should be 5,000 gram per square centimetre. We measure 4,935, 1.3 per cent lower because the discrete ray misses the centre. Halving the step changes the map by at most 0.24 per cent, and the no-void control is flat away from the chamber. Those two checks are what convince us the geometry is right before any statistics.

Slide 6 — Counts and reconstruction: the before/after image 1:25

On the slide.

- Raw counts: 191–1,197 per pixel, factor 6.3; the pyramid dominates.
- Twin target ratio: 1.0000 away, 1.4758 at the chamber.
- Peak: 366.8 with / 248.5 without; excess 118.3 muons in 30 days; 7.50σ.
- Before: density slice. After: clean ratio. Both are needed.

Figure / diagram. two large figures side by side: expected-count map and clean ratio map with profile. Point at the axes before interpreting.



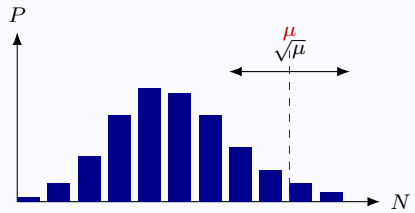
Spoken script. Step three gives counts, and step four gives the image. The raw 30-day map runs from 191 to 1,197 counts per pixel, a factor 6.3, because the pyramid is thicker in some directions. The chamber changes the total by only 0.36 per cent. So the raw map is a picture of the pyramid, not the chamber. We build a twin target with the chamber filled solid and divide. Away from the chamber the ratio is 1.0000 to four decimals. At the chamber it peaks at 1.4758: 366.8 counts with the chamber, 248.5 without, an excess of 118 muons in 30 days, which is 7.50 sigma expected. This ratio map is the after image. The density slice is the before image, and the gap between them is the whole point.

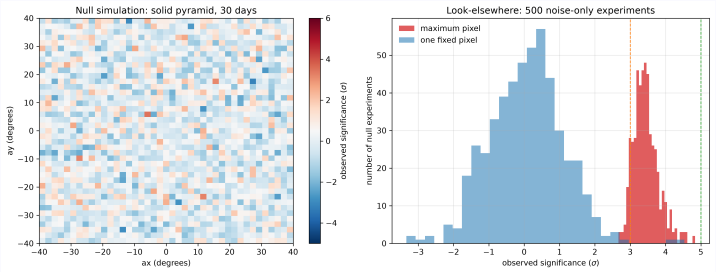
Slide 7 — Noise, significance, and the look-elsewhere effect 1:30

On the slide.

- Poisson: mean μ, spread $\sqrt{\mu}$.
- Observed $S_{\text{obs}} = (N - \mu_0)/\sqrt{\mu_0}$; expected $S_{\text{exp}} = (\mu_{\text{void}} - \mu_{\text{rock}})/\sqrt{\mu_{\text{rock}}}$.
- $S_{\text{exp}} \propto \sqrt{T}$; fix one signal pixel once.
- 1681 pixels: null maximum is 3.44σ on average; 91.6% of null maps have a 3σ pixel.

Figure / diagram. left: Poisson bar sketch; right: null ensemble figure with maximum and fixed-pixel distributions.





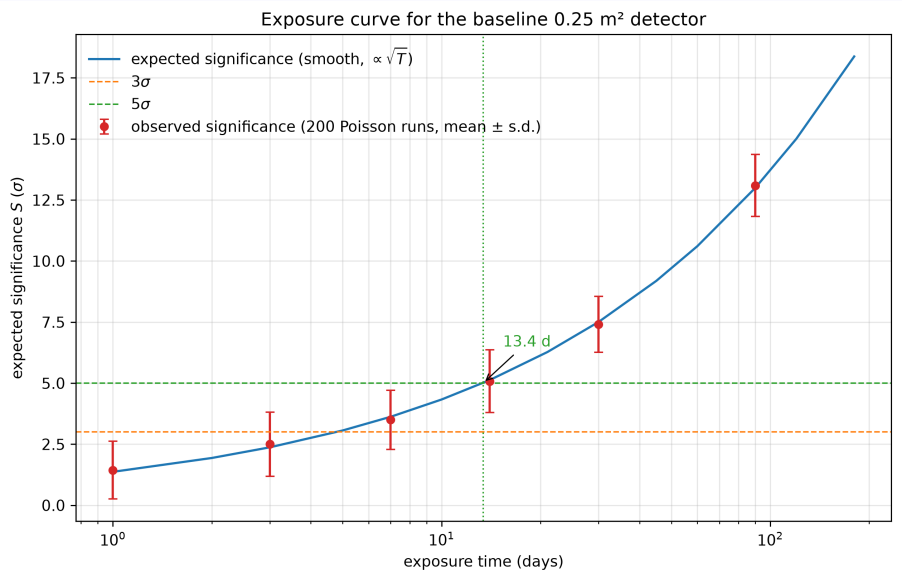
Spoken script. Now the statistics. A measurement is a Poisson draw: mean mu, spread root mu. Observed significance uses one noisy count and changes with the seed. Expected significance uses the two expectation maps and is deterministic. Because both signal and background grow with time, expected significance grows as root T. We choose one signal pixel from the noiseless maps and never move it. Why is that necessary? The null figure on the right: 500 solid-pyramid runs. One fixed pixel has mean zero and standard deviation one, as it should. The maximum over 1681 pixels has mean 3.44 sigma; 91.6 per cent of null maps contain a 3 sigma pixel. A single 3 sigma pixel is not a discovery.

Slide 8 — The exposure curve and the headline 1:30

On the slide.

- Fixed pixel: $S = 1.3696$ at 1 day; crosses 3σ at 4.82 days and 5σ at 13.4 days.
- 30 days: expected 7.50σ; observed seed-0 value 7.70σ.
- 200-run distributions: 14 days 5.12 ± 1.24 (54% cross); 30 days 7.53 ± 1.23 (98% cross).
- Deterministic benchmark, not a measurement.

Figure / diagram. large exposure curve with the $3\sigma/5\sigma$ lines and the observed points; a small definitions table in the corner.



Definition	30-day S	T_5 (days)
Peak pixel (official)	7.50σ	13.3
Exact pixel solid angle	6.98σ	15.4
Whole image (global)	3.55σ	59.5

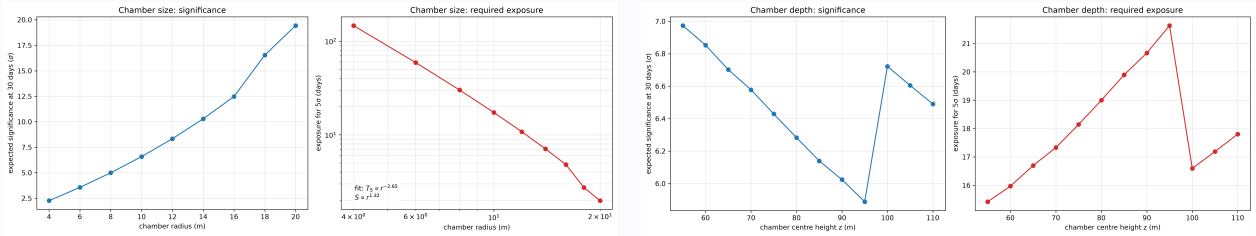
Spoken script. Here is the curve. At the fixed pixel, one day gives 1.3696 sigma. Three sigma comes at 4.82 days; five sigma at 13.4. Thirty days gives 7.50 expected, and our seed-zero observation gives 7.70. The red points are 200 runs at each exposure: at 14 days the mean is 5.12 and 54 per cent cross five sigma; at 30 days it is 7.53 and 98 per cent cross. The number is deterministic, so it has no error bar. The table shows the two alternative definitions: exact pixel solid angle gives 15.4 days, and whole-image global significance gives 59.5 days. Both are reported, but the official headline is the per-pixel expected value.

Slide 9 — Study A: size, and Study B: depth 1:25

On the slide.

- A: $r = 4$ m $\rightarrow$ 146.6 days; $r = 20$ m $\rightarrow$ 1.99 days; $S \propto r^{1.32}$.
- The signal is the chord ($\propto r$), not the volume ($\propto r^3$).
- B: depth 55–110 m; T_5 15.4 $\rightarrow$ 21.6 $\rightarrow$ 17.8 days.
- Shorter path competes with larger zenith angle; apex effect above 95 m.

Figure / diagram. two large figures side by side: Study A size scaling and Study B depth scaling.



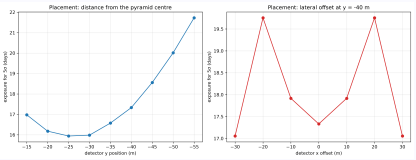
Spoken script. Study A asks how small a chamber we can find. From four to twenty metres the volume grows by 125, but the significance grows by only a factor nine and the exposure falls by about seventy. The fit is S proportional to r to the 1.3: the muon sees the missing chord, about two r, not the volume. Study B asks whether depth matters. The accessible range is 55 to 110 metres; below about 52 the chamber is outside the field of view. Exposure runs from 15.4 days at 55, up to 21.6 at 95, then down to 17.8 at 110. Less rock competes with a larger zenith angle; the rise near the apex is a pyramid geometry effect, not new physics.

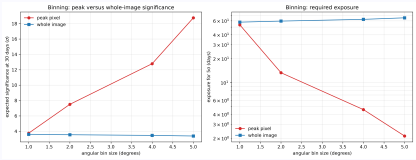
Slide 10 — Study C: placement, Study D: binning, and the hill 1:30

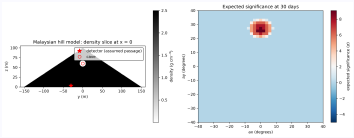
On the slide.

- C: best distance $y \approx -25$ m, 15.9 days; baseline 17.3; lateral flat within ± 30 m.
- D: peak S 3.75→18.75 as bins grow $1^\circ \to 5^\circ$; whole image stays $\approx 3.5\sigma$.
- D is a binning study, not a resolution study: no point-spread function.
- Malaysian hill: 8 m cave, 9.1 days.

Figure / diagram. three figures: C placement, D binning, and the Malaysian hill density/significance panel; keep the hill smaller.







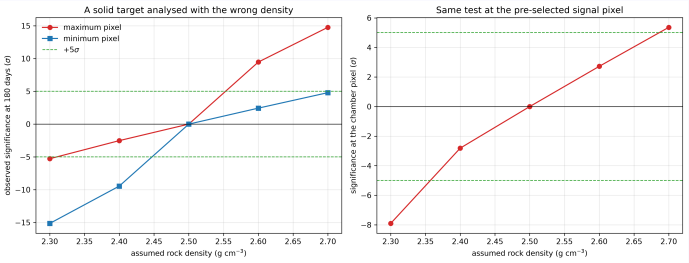
Spoken script. Study C is placement. The distance scan has an optimum near minus 25 metres, 15.9 days, against 17.3 at the baseline and 21.7 at minus 55. The lateral response is flat within 30 metres. Study D looks like a paradox: peak significance grows from 3.75 at one degree to 18.75 at five degrees. It is arithmetic: counts per pixel scale with solid angle, so signal over root background scales with bin size. The whole-image significance stays at about 3.5 sigma. A real detector would blur neighbouring directions, which we do not model, so this is a binning study, not a resolution study. Finally, the hypothetical Malaysian hill with an 8 metre cave needs about 9 days.

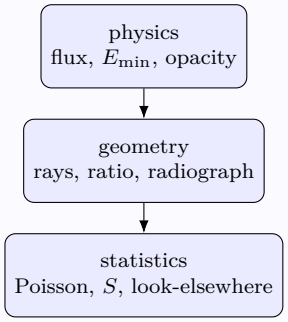
Slide 11 — Systematics, limitations, and conclusion 1:30

On the slide.

- Density: 4% error gives fake 3.86σ at 30 days and 9.46σ at 180 days; the deepest limitation.
- Detector efficiency/area: $T_5 \propto 1/(A\epsilon)$; 50% efficiency doubles exposure.
- Missing: scattering, backgrounds, atmospheric variation, real detector response, 3D tomography.
- Conclusion: 13.4 days for 5σ; geometry and statistics dominate.

Figure / diagram. left: density systematics; right: three-pillar TikZ summary (physics, geometry, statistics) with the headline underneath.





13.4 days

Spoken script. No result is better than its systematics. The density plot is the important one: if we assume the rock is four per cent denser than it is, a solid target produces a fake 3.86 sigma maximum at 30 days and 9.46 at 180, and a fake 2.72 sigma at the chamber pixel. A void and slightly lighter rock are degenerate. Detector area and efficiency scale the exposure inversely. We do not model scattering, backgrounds, atmospheric variation, or a real detector response, and we produce a radiograph, not a 3D inversion. The conclusion is still 13.4 days for five sigma under the stated model, controlled by geometry and statistics, not by the muon flux alone.

Slide 12 — Thank you, references, and Q&A 0:30

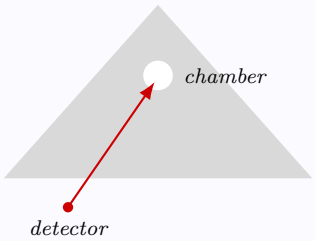
On the slide.

- Thank you. Questions welcome.
- Key references used on these slides:

[1] K. Morishima *et al.*, *Nature* **552**, 386 (2017).
[2] L. W. Alvarez *et al.*, *Science* **167**, 832 (1970).
[3] G. Bonomi *et al.*, *Prog. Part. Nucl. Phys.* **112**, 103768 (2020).
[4] Particle Data Group, *Phys. Rev. D* **110**, 030001 (2024).
[5] T. K. Gaisser, *Cosmic Rays and Particle Physics* (CUP, 1990).
[6] M. Guan *et al.*, arXiv:1509.06176 (2015).

Design note. *Render this list at 16–18 pt; short-form citations only. The full bibliography is in the paper.*

Figure / diagram. *a small TikZ pyramid with the detector and the words "13.4 days for 5σ" used as a closing logo.*



Spoken script. *Thank you. To summarise: muons give a free, penetrating beam; a twin-target ratio turns the pyramid into a chamber image; Poisson statistics and the look-elsewhere effect decide what counts as evidence; and the baseline model gives 13.4 days for five sigma. The main caveats are the rock density, the detector response, and the absence of a full 3D inversion. I am happy to take questions on the method, the numbers, or the limitations.*